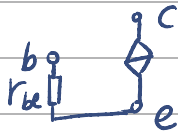


BJT

	A_v	R_i	R_o
CE 反相	> 100	$\times 10^2 \sim 10^3 \Omega$ 级	$R_C \times 10^2 \sim 10^3 \Omega$ 级
CC 同相	$\rightarrow 1$	$\times 10^1 \sim 10^2 k\Omega$ 级	$\times 10 \Omega$ 级
CB 同相	> 100	$\times 10^1 \Omega$ 级	$R_C \times 10^2 \sim 10^3 \Omega$ 级

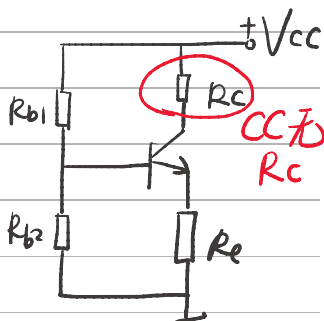
微变等效 BJT



直流模型

$$V_b = V_{CC} \cdot \frac{R_{b2}}{R_{b1} + R_{b2}}$$

$$r_b = \frac{R_{b1} R_{b2}}{R_{b1} + R_{b2}}$$



$$I_B r_b + V_{be} + (1 + \beta) I_B R_E = V_b$$

$$\Rightarrow I_B = \frac{V_b - V_{be}}{r_b + (1 + \beta) R_E}$$

可由 $I_C = \beta I_B$

$$R_C I_C + V_{CE} + (1 + \beta) I_B R_E = V_{CC}$$

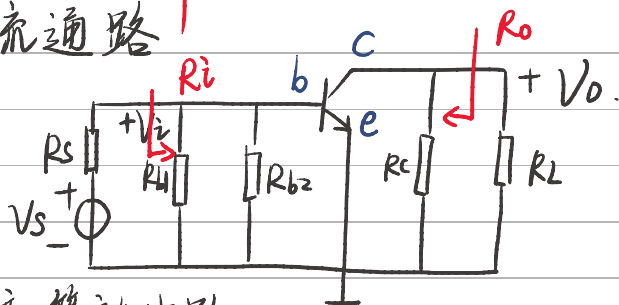
求得 V_{CE}

并根据 $r_{be} = r_{bb'} + \frac{V_T}{I_B}$ 可求 r_{be}

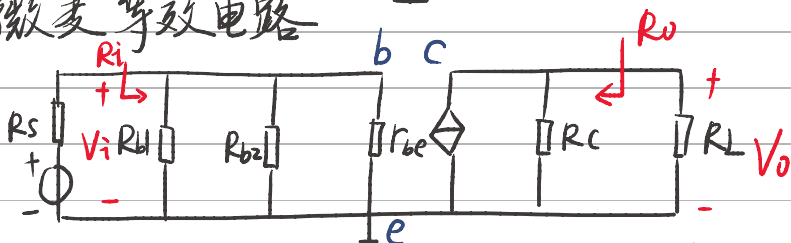
$r_{bb'}$ 以 200Ω 估计

CE 组态分析

交流通路



微变等效电路



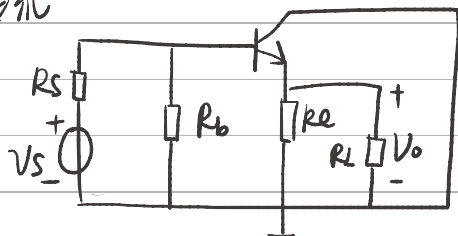
$$\dot{A}_V = \frac{\dot{V}_o}{\dot{V}_i} = - \frac{\beta \dot{I}_b (R_c \parallel R_L)}{\dot{I}_b r_{be}} \approx - \frac{\beta (R_c \parallel R_L)}{r_{be}}$$

$$R_i = \frac{\dot{V}_i}{\dot{I}_i} = R_{b1} \parallel R_{b2} \parallel r_{be}$$

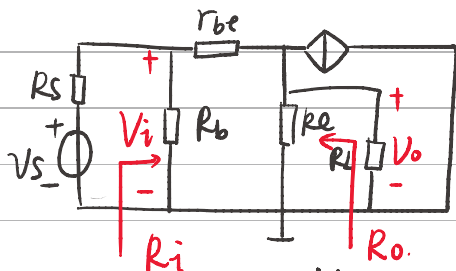
$$R_o = R_c$$

CC组态

交流



微变



$$R_L' = R_e \parallel R_L$$

$$\dot{A}_v = \frac{(1+\beta)I_b R_L'}{I_b r_{be} + (1+\beta)I_b R_L'} = \frac{(1+\beta)R_L'}{r_{be} + (1+\beta)R_L'} \approx 1$$

$$R_i = R_b \parallel R_i' \quad R_i' = \frac{V_i}{I_b} = r_{be} + (1+\beta)R_L'$$

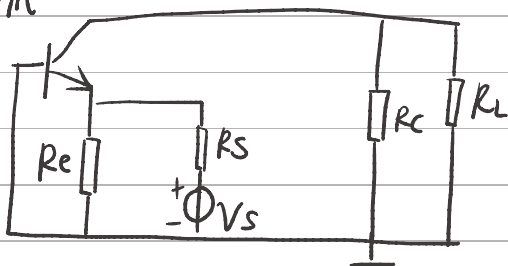
$$\Rightarrow R_i = R_b \parallel (r_{be} + (1+\beta)R_L')$$

$$R_o = R_e \parallel R_o' \quad R_o' = \frac{V_o'}{I_o'} = \frac{-I_b(r_{be} + R_b \parallel R_S)}{(1+\beta)I_b}$$

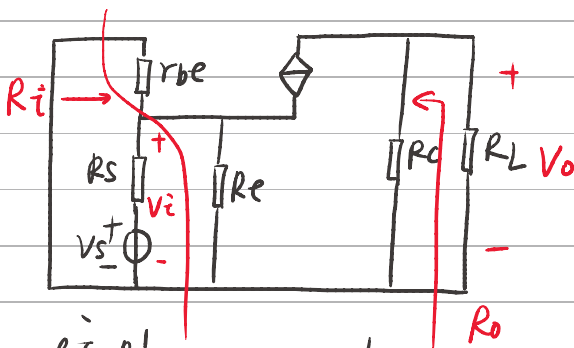
$$\Rightarrow R_o = R_e \parallel \frac{r_{be} + R_b \parallel R_S}{1+\beta}$$

CB 组态

交流



微变



$$\dot{A}_v = \frac{-\beta \dot{I}_b R_L'}{-\dot{I}_b r_{be}} = \frac{\beta R_L'}{r_{be}}$$

$$R_i = R_e \parallel R_i' \quad R_i' = \frac{\dot{V}_i}{-\dot{I}_e} = \frac{-\dot{I}_b r_{be}}{-(1+\beta)\dot{I}_b}$$

$$\Rightarrow R_i = R_e \parallel \frac{r_{be}}{1+\beta}$$

$$R_o = R_c$$

FET

	A_v	R_i	R_o
CS	反相 $\times 1 \sim 10$	M Ω 级	$\times 10^2 \sim 10^3 \Omega$ 级
CD	同相 $\rightarrow 1$	M Ω 级	$\times 10^3 \Omega$ 级
CG	同相 $\times 1 \sim 10$	$10^2 \Omega$ 级	$\times 10^2 \sim 10^3 \Omega$ 级

等效模型

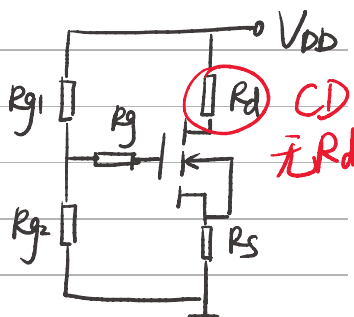


$$I_g = 0$$

直流模型

$$V_d = V_{DD} \cdot \frac{R_{g2}}{R_{g1} + R_{g2}}$$

$$R_d = R_g + \frac{R_{g1} R_{g2}}{R_{g1} + R_{g2}}$$



CD
无 R_d

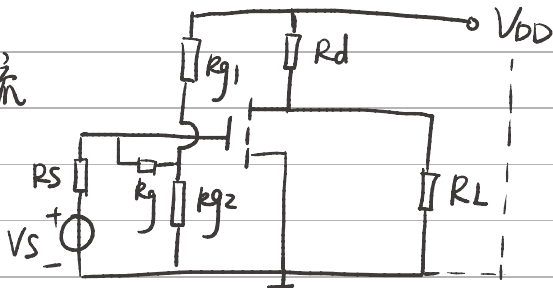
通过 I_D 表达式联立

$$I_D R_d + I_D R_s + V_{DS} = V_{DD}$$

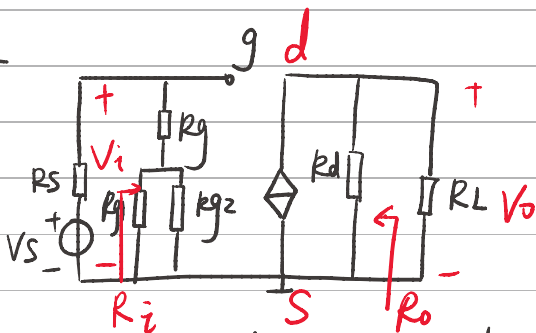
解得 V_{DS} I_D

CS

交流



微变



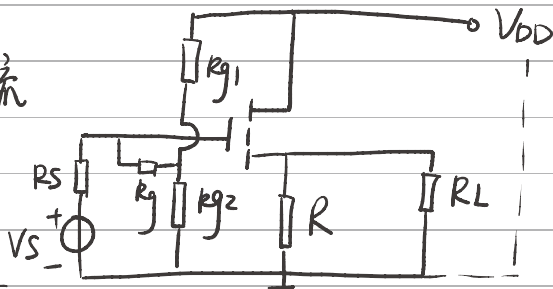
$$A_v = \frac{-g_m V_{gs} R_L'}{V_{gs}} = -g_m R_L'$$

$$R_i = R_g + R_{g1} \parallel R_{g2}$$

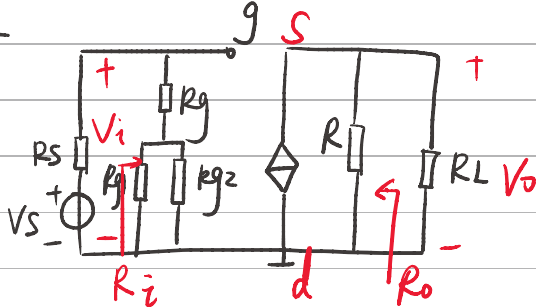
$$R_o = R_d \quad (V_S \text{ 置零})$$

CP

交流



微变



$$\dot{A}_v = \frac{g_m V_{gs} \cdot R_L'}{V_{gs} + g_m V_{gs} R_L'} = \frac{g_m R_L'}{1 + g_m R_L'} \approx 1$$

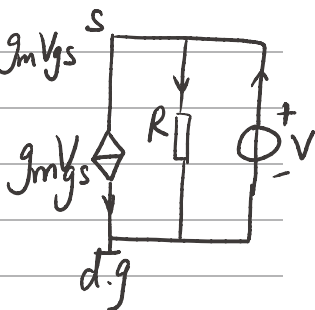
$$R_i = \frac{\dot{V}_i}{\dot{I}_i} = R_g + R_g \parallel R_{g2}$$

对 R_o 有 $\dot{V}_o = \dot{V}_{sd} = \dot{V}_{sg}$

$$I_R = \frac{-V_{gs}}{R} \quad \text{总电流 } I_o = -\frac{V_{gs}}{R} - g_m V_{gs}$$

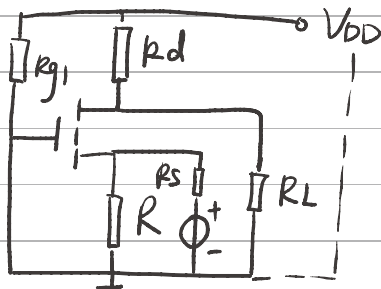
$$R_o = \frac{-V_{gs}}{-\frac{V_{gs}}{R} - g_m V_{gs}} = \frac{R \cdot \frac{1}{g_m}}{\frac{1}{g_m} + R}$$

$$= R \parallel \frac{1}{g_m}$$

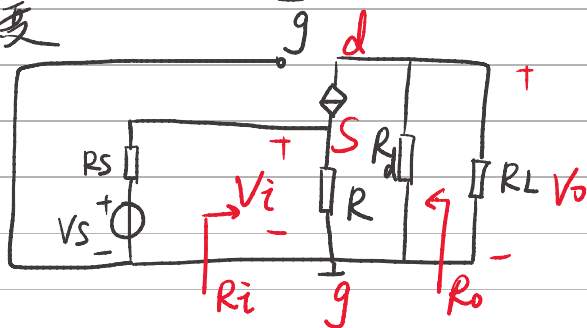


CS

交流



微变



$$\dot{A}_v = \frac{\dot{V}_o}{\dot{V}_i} = \frac{-g_m \dot{V}_{gs} R_L'}{-\dot{V}_{gs}} = g_m R_L'$$

$$R_i = R \parallel R' \quad R' = \frac{\dot{V}_i}{\dot{I}_d} = \frac{-\dot{V}_{gs}}{-g_m \dot{V}_{gs}} = \frac{1}{g_m}$$

$$\Rightarrow R_i = R \parallel \frac{1}{g_m}$$

$$R_o = R_d$$